

HAT-P-19b

R-1497

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_220830 · created 2026-08-01T14:23:24+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459821.8916 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.012
Transit depth (Rp/Rs)^2	3.34 +/- 0.45 [%]
Orbital Inclination (inc)	87.37 +/- 0.7
Airmass coefficient 1 (a1)	1.061 +/- 0.019
Airmass coefficient 2 (a2)	-0.054 +/- 0.065
Scatter in the residuals of the lightcurve fit is	1.83 %
Best Comparison Star	#3 - [536, 299]
Optimal Aperture	3.04
Optimal Annulus	5.81
Transit Duration (day)	0.104 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.012 vs 0.1418 (deviation 3.43σ)
Duration fit vs published	0.104 d vs 0.1196 d (deviation 1.2σ)
Mid-time O–C	-11.2 min
Depth SNR	15.2
Residual scatter	1.83 %

Light curve

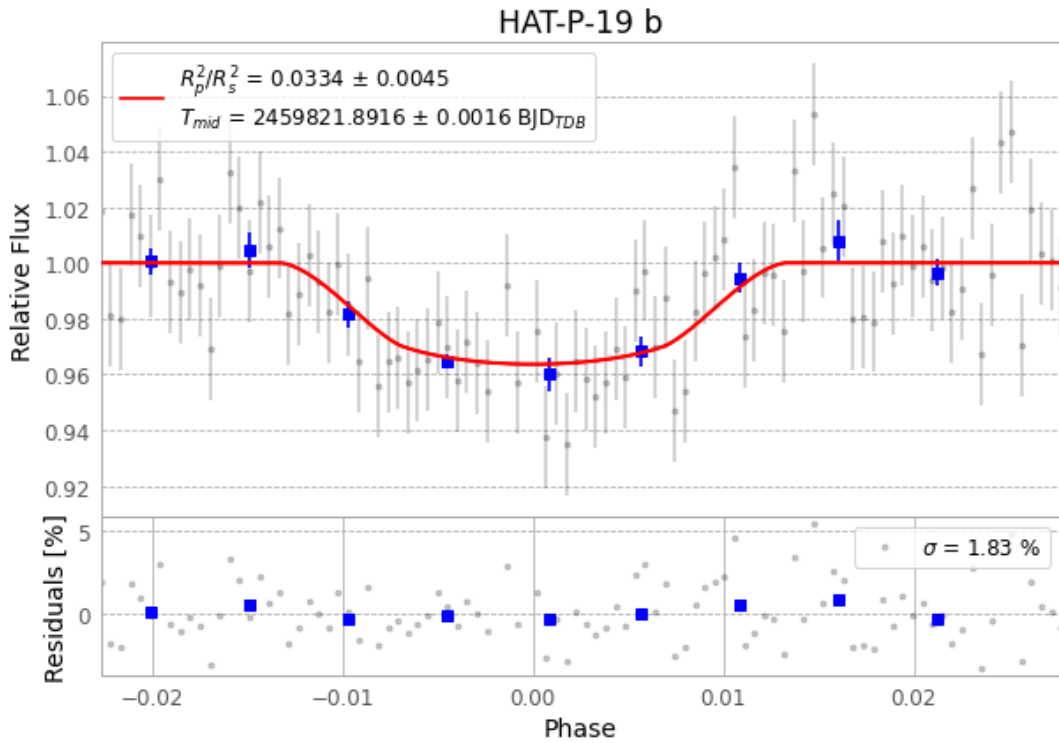


Figure 1 — FinalLightCurve_HAT-P-19 b_2022-08-30.png (SHA-256 e473a7f6002199f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [450, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf11fea25aca2bc2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), HATP-19220830070613.FITS → HATP-19220830115713.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-220830.log (5eb67ec50f3a...), FinalLightCurve_HAT-P-19 b_2022-08-30.png (e473a7f60021...), FinalLightCurve_HAT-P-19 b_2022-08-30.csv (46824c0e79b0...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 14:23Z.